

PARASEEDBENCH: WORDING, SEED, AND FINITE-GRID NONADDITIVITY IN TEXT-TO-IMAGE EVALUATION

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ABSTRACT

Text-to-image robustness depends jointly on prompt wording and sampled noise. ParaSeedBench crosses four meaning-equivalent wordings with eight shared seeds and exactly decomposes finite-grid semantic variation into wording, seed, and nonadditive wording–seed residual components. The main run contains 96 scenes and 11,520 images from three frozen 512×512 pipelines. Primary evidence is a prespecified two-rater audit of 1,536 images (16 complete grids) under neutral identifiers, with adjudication; full-suite automatic scores are secondary diagnostics. Inter-rater joint/atomic agreement was 0.969/0.948 ($\kappa = 0.937/0.891$). Within the audited pipelines, mean/all-four-wordings accuracy was 0.635/0.414 for PixArt- Σ , 0.357/0.156 for SD 1.5, and 0.332/0.133 for SDXL. On the same images, automatic scoring underestimated mean accuracy by 0.193 for PixArt- Σ , versus 0.080 and 0.076 for SD 1.5 and SDXL, demonstrating generator-dependent evaluator error. These are descriptive finite-grid quantities; one image per cell does not identify a noise-corrected population interaction. The release provides frozen prompts, seeds, provenance, and auditable labels.

Index Terms— text-to-image generation, robustness, semantic evaluation, experimental design, reproducibility

1. INTRODUCTION

Text-to-image (T2I) generators are stochastic systems: an image depends on both a linguistic conditioning signal and sampled noise. Their evaluation should distinguish variation in requested content from permissible variation in unspecified appearance. Given “a cat left of a dog,” a change of background is not necessarily an error, but reversing the relation is. Conversely, consistently omitting the dog is stable failure, not reliable generation.

A common diagnostic fixes the seed and compares images generated from equivalent prompts. Another holds the prompt fixed and changes seeds. These comparisons are useful, but neither alone attributes variation exclusively to wording or noise. A generator may react to a wording only for particular initializations. This nonadditivity affects both comparisons. Moreover, equality of output distributions does not imply point-wise equality under a chosen noise coupling: for symmetric Z ,

the variables Z and $-Z$ have identical distributions although paired realizations differ.

We study a narrow measurement question: *within a specified prompt–seed grid, how much observed semantic variation is associated with wording, seed, and their nonadditive residual, and how does each coexist with correctness?* The contribution is an auditable evaluation design, not a new generator or a claim that paraphrase testing itself is novel. We combine an exact descriptive decomposition, accuracy diagnostics, a controlled prompt suite, and a checkpointed implementation with explicit provenance and completeness checks. The human audit further shows why automatic correctness cannot simply be treated as ground truth. Generalization beyond the tested templates, seeds, and pipelines requires further evidence.

2. RELATIONSHIP TO PRIOR EVALUATION

TIFA evaluates faithfulness through visual question answering [1]; GenEval evaluates compositional object, count, color, and spatial requirements [2]. SemVarBench examines responses to linguistic permutations [3]. MetaLogic directly tests logically equivalent prompts and diagnoses object, count, and positional inconsistencies [4]. Thus, equivalent-prompt evaluation and these semantic categories are established ideas. Studies of reliable seeds also demonstrate that initialization can affect compositional performance [5].

Unlike reliable-seed studies that characterize or optimize initialization and paraphrase benchmarks that vary language, ParaSeedBench jointly crosses both axes and validates complete-grid outcomes through independent human labeling and adjudication. Its incremental contribution is a *crossed, correctness-aware analysis* of wording, seed, and their nonadditive residual. It uses standard finite-table orthogonal decomposition, not a new analysis-of-variance estimator. GenEval 2 documents evaluator drift [6], motivating validation of the semantic response grid. We therefore make adjudicated human states primary on a prespecified main subset and retain the automatic evaluator as a secondary diagnostic.

3. FINITE-GRID SEMANTIC ANALYSIS

3.1. Experimental unit and correctness

Scene c specifies K_c binary visual requirements and J equivalent wordings p_{cj} . For each model separately,

$$x_{cjs} = G(p_{cj}, z_s), \quad \mathbf{q}_{cjs}^H, \mathbf{q}_{cjs}^M \in \{0, 1\}^{K_c}. \quad (1)$$

Here H denotes the resolved human state and M the OWLv2/CLIP state. The seed is reset for every call; wordings share a random stream within a model. Matching integer seeds across architectures does not align their latent semantics. Unless superscripted, the equations below apply to either label source; confirmatory semantic results use H , while M is secondary. Joint correctness is $A_{cjs} = \prod_k q_{cjsk}$.

Two primary descriptive endpoints are mean accuracy and empirical worst-wording accuracy:

$$\bar{A} = \frac{1}{CJS} \sum_{cjs} A_{cjs}, \quad R_{\text{emp}} = \frac{1}{C} \sum_c \min_j \frac{1}{S} \sum_s A_{cjs}. \quad (2)$$

R_{emp} is the minimum on a finite observed set, not guaranteed robustness. Even if all wordings have the same true success rate, selecting the lowest sample mean induces downward bias. A scene bootstrap does not remove it. As a sensitivity diagnostic, consider all 35 unordered balanced splits of eight seeds. For each split, select every wording tied for lowest training-half accuracy, average their held-out scores, reverse the halves, and then average across splits. This held-out selected-worst score R_{ho} is invariant to wording order and seed ordering, but remains a sensitivity diagnostic rather than an unbiased population minimum. We also report the fraction of seeds for which *all* four wordings are correct; this is a different estimand.

3.2. Wording, seed, and nonadditivity

Suppress c , define $\|v\|_K^2 = K_c^{-1} \sum_k v_k^2$, and let

$$\begin{aligned} \mu &= \frac{1}{JS} \sum_{js} \mathbf{q}_{js}, & a_j &= \frac{1}{S} \sum_s \mathbf{q}_{js} - \mu, \\ b_s &= \frac{1}{J} \sum_j \mathbf{q}_{js} - \mu, & r_{js} &= \mathbf{q}_{js} - \mu - a_j - b_s. \end{aligned} \quad (3)$$

The finite-grid variation components are

$$\begin{aligned} V_w &= \frac{1}{J} \sum_j \|\mathbf{a}_j\|_K^2, & V_s &= \frac{1}{S} \sum_s \|\mathbf{b}_s\|_K^2, \\ V_{ws} &= \frac{1}{JS} \sum_{js} \|\mathbf{r}_{js}\|_K^2. \end{aligned} \quad (4)$$

Their sum equals total variation:

$$\frac{1}{JS} \sum_{js} \|\mathbf{q}_{js} - \mu\|_K^2 = V_w + V_s + V_{ws}. \quad (5)$$

Table 1. Analytical one-atom 2×2 examples, not model measurements. Entries of q are arranged as wording rows and seed columns.

q	$\bar{A}$	V_w	V_s	V_{ws}	(D_w, D_s)
(0, 0; 0, 0)	0	0	0	0	(0, 0)
(1, 1; 1, 1)	1	0	0	0	(0, 0)
(0, 0; 1, 1)	.5	.25	0	0	(1, 0)
(0, 1; 1, 0)	.5	0	0	.25	(1, 1)

To see this, expand the square. Each row and column mean of r is zero, and $\sum_j a_j = \sum_s b_s = 0$; all cross terms vanish. These are exact empirical components, not unbiased population variance estimates. With one image per cell, numerical nondeterminism and judge error cannot be separately estimated as residual noise. For $J = 4, S = 8$, an iid Bernoulli no-effects reference assigns expected nonconstant-grid shares $3/31, 7/31$, and $21/31$ to V_w, V_s, V_{ws} , respectively, from their ranks. Raw V_{ws} shares therefore reflect the residual’s 21 degrees of freedom and are not evidence that a latent interaction dominates noise.

For binary states, normalized Hamming distance equals squared distance under $\|\cdot\|_K^2$. Define D_w as the mean Hamming distance across unordered wording pairs at each seed, and D_s across unordered seed pairs at each wording. The pairwise-distance identity gives

$$D_w = \frac{2J}{J-1}(V_w + V_{ws}), \quad D_s = \frac{2S}{S-1}(V_s + V_{ws}). \quad (6)$$

Thus the nonadditive residual contributes to both; comparing their raw magnitudes also involves different finite-size factors. All components are first computed per scene, then averaged with equal category weights. Low variation is interpreted jointly with accuracy, never as success on its own.

4. VERSIONED BENCHMARK PROTOCOL

4.1. Scenes and data boundaries

Version 2 contains 96 scenes, 24 per category: object existence, exact count, color binding, and spatial relation. Each has four wordings and one meaning-changing control. Two cyclic pairings over 12 concrete nouns replace lexicographically truncated pairs. Counts two, three, and four each appear in eight scenes. All variants request a photograph; spatial paraphrases preserve relative position without introducing sitting or absolute image-edge constraints. Color and spatial prompts explicitly require exactly one instance of each queried object, resolving the evaluator’s highest-confidence-instance ambiguity. Other unrequested objects are allowed.

There are 16 separate development scenes using four additional object nouns; a four-scene smoke subset is drawn from these, not from the main set. Main and development semantic specifications are disjoint, including controls. Different

development nouns introduce a calibration-domain limitation. Within the main set, count controls can reproduce another base count, and shared templates limit population-level interpretation. The development audit did not justify evaluator calibration, so thresholds remain declared defaults. Main images did not guide prompts or thresholds. The main prompt set and execution contract were frozen before the completed run; the release preserves the prompt specifications, locked run provenance, and selection manifest.

4.2. Generators and execution contract

The controlled 512×512 study includes Stable Diffusion 1.5 [7], SDXL-base [8], and PixArt- Σ [9], with respectively 30/30/20 steps and guidance 7.5/7.5/4.5. PixArt uses the 512-MS denoising transformer with the T5, VAE, tokenizer, and scheduler components from its complete 1024-MS Diffusers pipeline; both repository commits are pinned. The scheduler configuration is saved. These settings define operational pipelines, not compute-matched or best-quality model comparisons. In particular SDXL’s lower-than-preferred resolution requires a separate native-resolution sensitivity experiment before broader ranking claims.

The seed list is [11, 29, 47, 71, 101, 131, 173, 211], shared across wordings. A fresh CPU generator is created for every call. FP16 generation, batch size one, disabled TF32, deterministic PyTorch algorithms, fixed negative prompt, and immutable checkpoint commits define the run. PixArt caption cleaning is disabled explicitly. Model-native safety filtering is retained; flagged and all-black outputs are recorded separately and assigned zero correctness for *end-to-end pipeline* accuracy. The uncensored subset is reported only descriptively. Rerolling flagged seeds or silently discarding them would change the sampling design.

The main grid contains $96 \times 5 \times 8 = 3840$ images per model, or 11,520 total. The RTX 6000 Ada profile keeps models resident on GPU; the RTX 3070 profile offloads components to CPU. These are separate execution profiles. Neither runtime nor cross-platform numerical equivalence is assumed. Package versions, source hash, GPU model, CUDA/cuDNN, elapsed generation time, and peak allocated memory are recorded. Resumption rejects changed run contracts. A same-process duplicate-input check covers one base scene per category and model; it does not establish cross-device or cross-process determinism.

4.3. Secondary automatic evaluator and controls

The evaluator uses OWLv2-base-patch16-ensemble [10] and CLIP ViT-B/32 [11] in FP32, pinned to immutable revisions. OWLv2 uses an explicitly selected slow processor rather than a version-dependent default. Thresholds 0.10 for detection and 0.35 for class-wise NMS are declared defaults, not validated optima. Color is classified among ten alternatives using

the highest-scoring object’s box with 3% padding. Spatial predicates use centroids with a 0.03 image-width/height dead zone. Missing objects fail dependent atoms; exact-one atoms fail when extra queried instances are detected. Correlated detection, color, and boundary errors motivate its secondary status.

For count, color, and spatial controls, evaluate each base/control image against *both* specifications. A paired discrimination success requires each image to satisfy its own specification and fail the alternative. Presence replacement is non-exclusive: an image can satisfy both prompts. It therefore receives own-prompt accuracy but no exclusive-discrimination score. All controls are excluded from the main variation decomposition.

5. HUMAN-PRIMARY MAIN RESULTS

5.1. Resolved two-rater audit

Using seed 2027, the audit selected four scenes uniformly without replacement within each category, independently of scores. All four equivalent wordings, eight seeds, and three models gave 1,536 images. Two raters independently labeled every semantic atom under neutral identifiers. Both rating interfaces withheld model identity and automatic labels; the adjudication interface also withheld the two prior state vectors, although visual style could still reveal model family. A separate adjudication pass resolved disagreements and was not treated as a third independent rating. Before adjudication, 144 *images* had unequal state vectors, comprising 199 differing atom bits (141 versus 58 in the two directions). Inter-rater atomic agreement was 0.948 with $\kappa = 0.891$; joint correctness agreement was 0.969 with $\kappa = 0.937$.

Under the frozen 512×512 pipelines, Table 2 gives audited mean accuracy 0.635 for PixArt- Σ , 0.357 for SD 1.5, and 0.332 for SDXL. Paired scene differences were +0.277 [0.197,0.365] for PixArt versus SD 1.5, +0.303 [0.232,0.373] for PixArt versus SDXL, and +0.025 [−0.033,0.084] for SD 1.5 versus SDXL. These operational comparisons are not architecture rankings. Stability alone did not imply correctness.

The split-symmetrized human-primary R_{ho} values were 0.580 [0.517,0.655] for PixArt- Σ , 0.315 [0.245,0.383] for SDXL, and 0.347 [0.248,0.432] for SD 1.5. For the variance components, intervals were: PixArt- Σ , V_w [.007,.014], V_s [.026,.048], V_{ws} [.044,.061]; SDXL, [.011,.022], [.074,.106], [.071,.094]; and SD 1.5, [.012,.026], [.049,.072], [.070,.096]. Dividing each H16 component by its rank gives, in 10^{-3} units, wording/seed/residual estimates of 3.3/5.3/2.5 for PixArt- Σ , 5.3/12.9/4.0 for SDXL, and 6.0/8.7/4.0 for SD 1.5. The seed estimate was largest for every pipeline; the reported marginal intervals separate it from wording only for SDXL. V_{ws} is the realized nonadditive residual and includes unresolved cell-level variation.

Table 2. Human-primary results on the prespecified 16-scene main audit. Brackets are descriptive 95% category-stratified scene-bootstrap intervals (2,000 replicates).

Metric	PixArt- Σ	SDXL	SD 1.5
Mean accuracy	0.635 [0.562, 0.705]	0.332 [0.287, 0.375]	0.357 [0.289, 0.418]
Empirical worst accuracy	0.484 [0.422, 0.555]	0.203 [0.141, 0.258]	0.234 [0.148, 0.312]
All-wordings accuracy	0.414 [0.344, 0.492]	0.133 [0.094, 0.172]	0.156 [0.078, 0.219]
Wording disagreement	0.166 [0.136, 0.198]	0.265 [0.223, 0.299]	0.271 [0.227, 0.318]
Seed disagreement	0.204 [0.169, 0.238]	0.395 [0.365, 0.427]	0.328 [0.286, 0.373]
V_{ws}	0.052 [0.044, 0.061]	0.083 [0.071, 0.094]	0.083 [0.070, 0.096]

Table 3. Finite-grid variance components. H16 is human-primary on the 16-scene audit; M16 is automatic scoring on the same grids; M96 is automatic scoring on the full 96-scene main suite.

Scope	Model	V_w	V_s	V_{ws}
H16	PixArt- Σ	.010	.037	.052
H16	SDXL	.016	.090	.083
H16	SD 1.5	.018	.061	.083
M16	PixArt- Σ	.012	.046	.064
M16	SDXL	.012	.061	.072
M16	SD 1.5	.018	.056	.071
M96	PixArt- Σ	.014	.045	.061
M96	SDXL	.015	.073	.072
M96	SD 1.5	.023	.058	.080

5.2. Automatic diagnostics and evaluator shift

The full main automatic run is complete: 3,840 images per model, 11,520 total. Its M96 values in Table 3 are descriptive secondary diagnostics. On the audit, machine-human joint accuracy/ κ was 0.725/0.466 for PixArt- Σ , 0.850/0.640 for SDXL, and 0.834/0.620 for SD 1.5; atomic accuracy was 0.813, 0.817, and 0.804. Machine-minus-human mean accuracy was -0.193 , -0.076 , and -0.080 , respectively. This generator-dependent error precludes treating M96 as confirmatory ground truth. Paired discrimination requires both images to satisfy their own specification and fail the alternative. Across $72 \times 8 = 576$ eligible pairs per model, all four conditions held for 37 PixArt- Σ , 5 SDXL, and 18 SD 1.5 pairs (0.064, 0.009, and 0.031); this is an end-to-end stress test, not evaluator calibration.

Automatic generation retained all model-native safety outcomes. SD 1.5 produced 19 censored/all-black outputs; SDXL and PixArt- Σ produced none. Median measured generation times on the RTX 6000 Ada were 0.825, 1.506, and 0.877 s/image for SD 1.5, SDXL, and PixArt- Σ , respectively. Run-time is descriptive because the pipelines are not compute-matched.

5.3. Uncertainty and interpretation

Intervals use 2,000 scene-bootstrap replicates within category with equal category weights. With only four audited scenes per category, percentile coverage is uncertain; we treat them

as descriptive stability intervals under resampling of the selected scenes. They condition on the four wording templates, eight seeds, pipelines, and label protocol, do not cover unseen prompt families or seeds, and do not remove empirical-minimum selection bias. No multiplicity-adjusted hypothesis tests are claimed.

6. LIMITATIONS AND CONCLUSION

The suite measures simple, templated photographic compositions rather than unrestricted intent. Meaning equivalence was author-designed but not independently validated. Components depend on the selected wording/seed set, and one observation per cell precludes separating latent interaction from residual variation. Primary labels retain human subjectivity; adjudication was not a third independent rating. The audit samples only 16 scenes. Category comparisons of joint accuracy are additionally entangled with one to four required atoms per scene. Safety behavior differs by pipeline, and equal output resolution is not equal native capability. No claim of novelty over existing paraphrase benchmarks follows from dataset size or a crossed table alone.

ParaSeedBench partitions observed semantic variation on a crossed wording-seed grid while keeping correctness explicit. The human-primary audit shows that evaluator choice can materially shift both success rates and apparent instability. The benchmark therefore treats adjudicated human states as primary evidence and full-suite automatic scoring as a reproducible secondary diagnostic. The release supplies frozen prompts, seeds, provenance checks, audited labels, and restoration tools.

7. COMPLIANCE WITH ETHICAL STANDARDS

OpenAI ChatGPT assisted with language editing, L^AT_EX preparation, and portions of release-code drafting and review. It was neither an evaluated generator nor used to produce or adjudicate human labels. The authors designed and ran the study, verified the code, analyses, results, and citations, and take responsibility for the manuscript.

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